AWS SAA-C03 Study Note: Databases

1. Choosing the Right Database (Exam Cheat Sheet)

The exam often presents a scenario and asks for the *best* database. Use this keyword mapping:

Scenario / Keywords	Recommended Service
Relational (SQL), OLTP, traditional apps (ERP, CRM)	**RDS**
Relational, **High Performance**, **Auto-scaling storage**, Enterprise-grade	**Aurora**
NoSQL, Key-Value, High Scale, **Single-digit millisecond latency**	**DynamoDB**
In-memory cache, **Sub-millisecond latency**, alleviate DB load	**ElastiCache**
Graph data, Social networks, Recommendation engines	**Neptune**
MongoDB compatibility, JSON documents	**DocumentDB**
Cassandra migration, Wide-column store	**Keyspaces**
Time-series data, IoT streams, DevOps logs	**Timestream**
Data Lake, Unstructured data, Static website hosting	**S3**
Data Warehousing, OLAP, SQL Analytics	**Redshift** (Note: distinct from RDS)

2. Amazon RDS (Relational Database Service)

Managed service for DBs use SQL as a query language.

- **Engines:** Postgres, MySQL, MariaDB, Oracle, Microsoft SQL Server.
- **Key Features:**
  - **Multi-AZ (Availability Zones):**
    - **Purpose: Disaster Recovery (High Availability)**.
    - Synchronous replication to a standby instance in a different AZ.
    - Automatic failover if the primary goes down (CNAME record flips).
    - *Not* for scaling performance.
  - **Read Replicas:**
    - **Purpose: Scalability (Performance)**.
    - Asynchronous replication.
    - Can be within AZ, Cross-AZ, or Cross-Region.

        - Apps must update the connection string to read from replicas.
    - **Backups:**
        - **Automated Backups:** Daily full backup + transaction logs (5-minute point-in-time recovery). Retention 1-35 days.
        - **Snapshots:** Manual, user-initiated, stored indefinitely.
    - **Storage:**
        - **General Purpose SSD (gp2/gp3):** Cost-effective, burstable.
        - **Provisioned IOPS (io1/io2):** For high I/O, latency-sensitive workloads.
    - **Encryption:** Can strictly only be enabled at creation time. To encrypt an unencrypted DB, you must snapshot it -> copy snapshot (encrypting it) -> restore from encrypted snapshot.

3. Amazon Aurora

AWS-proprietary relational database compatible with MySQL and Postgres. 5x faster than MySQL, 3x faster than Postgres.

- **Architecture:**
    - **Separation of Compute and Storage.**
    - **Storage:** Data is replicated **6 times across 3 AZs**. Self-healing.
    - Storage auto-expands (no need to provision storage size manually).
- **Endpoints:**
    - **Cluster Endpoint:** Points to the *Writer* instance.
    - **Reader Endpoint:** Load balances connections across all *Read Replicas*.
- **High Availability:**
    - Can have up to **15 Read Replicas** (RDS limit is 5).
    - Failover is faster than RDS (usually < 30 seconds).
- **Aurora Serverless:**
    - Auto-scales compute based on demand.
    - Good for: Infrequent, intermittent, or unpredictable workloads.
- **Aurora Global Database:**
    - One primary region (R/W), up to 5 secondary read-only regions.
    - Latency < 1 second across regions. Promotes secondary to primary in < 1 minute.

4. Amazon ElastiCache

Managed Redis or Memcached. In-memory caching to improve read performance.

- **Use Cases:** Reduce load on RDS/DynamoDB, store user sessions, leaderboards.
- **Redis vs. Memcached (Crucial for Exam):**

Feature	Redis	Memcached
Data Structure	Complex (Sets, Lists, Sorted Sets)	Simple String/Object
Persistence	Yes (can save to disk)	No (Purely in-memory)
Availability	Multi-AZ, Failover, Replication	No Replication
Scaling	Scale Read Replicas	Multi-thread architecture (Scale Up)
Exam Tip	Choose if you need **High Availability**, **Backup**, or **Sorting**.	Choose for **simple** caching or **multithreaded** performance.

- **Caching Strategies:**
  - **Lazy Loading:** Load data only when requested. (Pros: Only requested data cached. Cons: Cache miss penalty).
  - **Write-Through:** Add data to cache whenever DB is updated. (Pros: Data never stale. Cons: Write penalty, potentially unused data).

5. Amazon DynamoDB

Fully managed, Serverless, NoSQL Key-Value store.

- **Performance:** Single-digit millisecond latency at any scale.
- **Capacity Modes:**
  - **Provisioned:** You set RCU (Read Capacity Units) and WCU (Write Capacity Units). Use when traffic is predictable.
  - **On-Demand:** Auto-scales instantly. Use for unpredictable/spiky traffic.
- **Keys:**
  - **Partition Key (Hash):** Determines data distribution.
  - **Sort Key (Range):** Sorts data within the partition.
- **DAX (DynamoDB Accelerator):**
  - Fully managed, in-memory cache for DynamoDB.
  - **Keyword:** Improves latency from milliseconds to **microseconds**.
  - Use when you have a "hot key" problem or extreme read loads.
- **DynamoDB Streams:**
  - Captures item-level changes (INSERT, UPDATE, DELETE).
  - Triggers Lambda functions (e.g., "When a new user signs up, send a welcome email").
- **Global Tables:**
  - Multi-Region, Active-Active replication.
  - High availability and local reads/writes in multiple regions.

6. Amazon S3 (As a Database)

While essentially Object Storage, S3 is often a component in database architectures.

- **Data Lake:** S3 acts as the central repository for structured and unstructured data.
- **Query in Place:**
  - **Amazon Athena:** Serverless SQL queries directly on S3 data.
  - **S3 Select:** Retrieve only a subset of data from an object (improves performance).
- **Consistency:** Strong consistency for all GETs, PUTs, and LISTs.

7. Amazon DocumentDB (with MongoDB compatibility)

- **Keyword: MongoDB**.
- Used for managing JSON data.
- Architecture similar to Aurora (storage replicates 6 ways across 3 AZs).

8. Amazon Neptune

- **Keyword: Graph Database**.
- **Use Cases:**
  - Social Networking (Who is friends with whom?).
  - Recommendation Engines (If you bought A, you might like B).
  - Fraud Detection (Pattern matching).
  - Knowledge Graphs.
- Query Languages: Gremlin (Apache TinkerPop), SPARQL.

9. Amazon Keyspaces (for Apache Cassandra)

- **Keyword: Cassandra**.
- Serverless, scalable, highly available wide-column store.
- Compatible with CQL (Cassandra Query Language).
- Use this if migration questions mention existing Cassandra workloads.

10. Amazon Timestream

- **Keyword: Time-series**.
- **Use Cases:**
  - IoT applications (sensor readings every second).
  - DevOps (tracking CPU/Memory usage over time).
  - Industrial telemetry.
- Architecture:
  - **Memory Store:** For recent data (fast ingestion).
  - **Magnetic Store:** For historical data (cost-optimized).

- Built-in retention policies to move data between stores automatically.